

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month, in Earth we take a look at the issue of nutrient pollution, in Health, we continue to highlight research and work towards fighting the pandemic, in Ignorance, we look at insect wing evolution, and much more.



New Allosaurus discovered

A new species, *Allosaurus jimmadseni*, has been described based on two well preserved and nearly complete skeletons. The new species is 155 million years ago, and is the oldest known *Allosaurus*. The species has a slender narrow skull, and was the top predator in its ecosystem. <http://dgit.in/newallo>

WHAT'S NEW

NASA funds research into autonomous air cargo transportation

The rollout of the vaccines for Covid-19 has resulted in the largest global logistics effort since World War II. There are all kinds of new stresses on the supply chain that the current logistical setups are just not prepared to deal with. The pandemic has highlighted the criticality of reducing the inefficiencies in supply chains, and autonomous air cargo has been envisioned as technology with a promise of doing just that. The researchers will be providing both the theory and concept for all kinds of aerial delivery vehicles, from large autonomous blimps ferrying containers, to tiny urban drones making



home deliveries in residential neighbourhoods. The researchers will be addressing the long list of concerns and challenges, including

management of autonomous aircraft traffic flow, regulation, safety, privacy, emissions and noise pollution. To find solutions for these issues, the researchers will rely heavily on computational engineering and machine learning.

<http://dgit.in/ftrcrgeo>

Dancing robots confirm theory on order in active matter

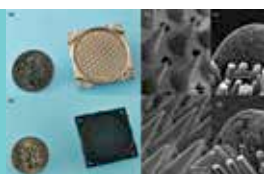
Researchers have identified for a new principle by which order can emerge in the chaotic and diverse collections of "particles", or active matter systems. The researchers describe new states of active matter systems, as high rattling and low rattling. In the high rattling state a lot of energy is used up and converted by the particles into violent and



random motions. In a low rattling state, the motions are slight and highly organised. The theory was that if matter and energy allow for a low rattling state, the active matter

system would randomly rearrange until that state is found, where it would then settle. The researchers used a swarm of robots (known as smarticles) to confirm the theory, and found that the robots indeed settled into a low rattling state, and began to dance, the choreography emerging from the physics of low rattling states.

<http://dgit.in/lowrat>



Nanosat thruster

Researchers from MIT have created the first of its kind nanosatellite thruster that produces pure ions, with the entire component being additively manufactured. The thruster uses pure ions to provide a tiny amount of force, less than half the weight of a sesame seed on top of a burger. <http://dgit.in/nnothrstr>



Striped brown dwarf

Astronomers wanted to find out if the nearest brown dwarf, a failed star somewhere between a gas giant such as Jupiter and a red dwarf star had spots and storms on its surface or stripes. They found stripes, which indicates an atmosphere dominated by high speed winds. <http://dgit.in/strdwrf>



Juno mission extended

This is really great news because the citizen science objective of the mission means that the imaging community can continue to use the data from the Juno mission to produce even more spectacular images of Jupiter. The extension is till 2025, so more science can get done as well. <http://dgit.in/junomis>